

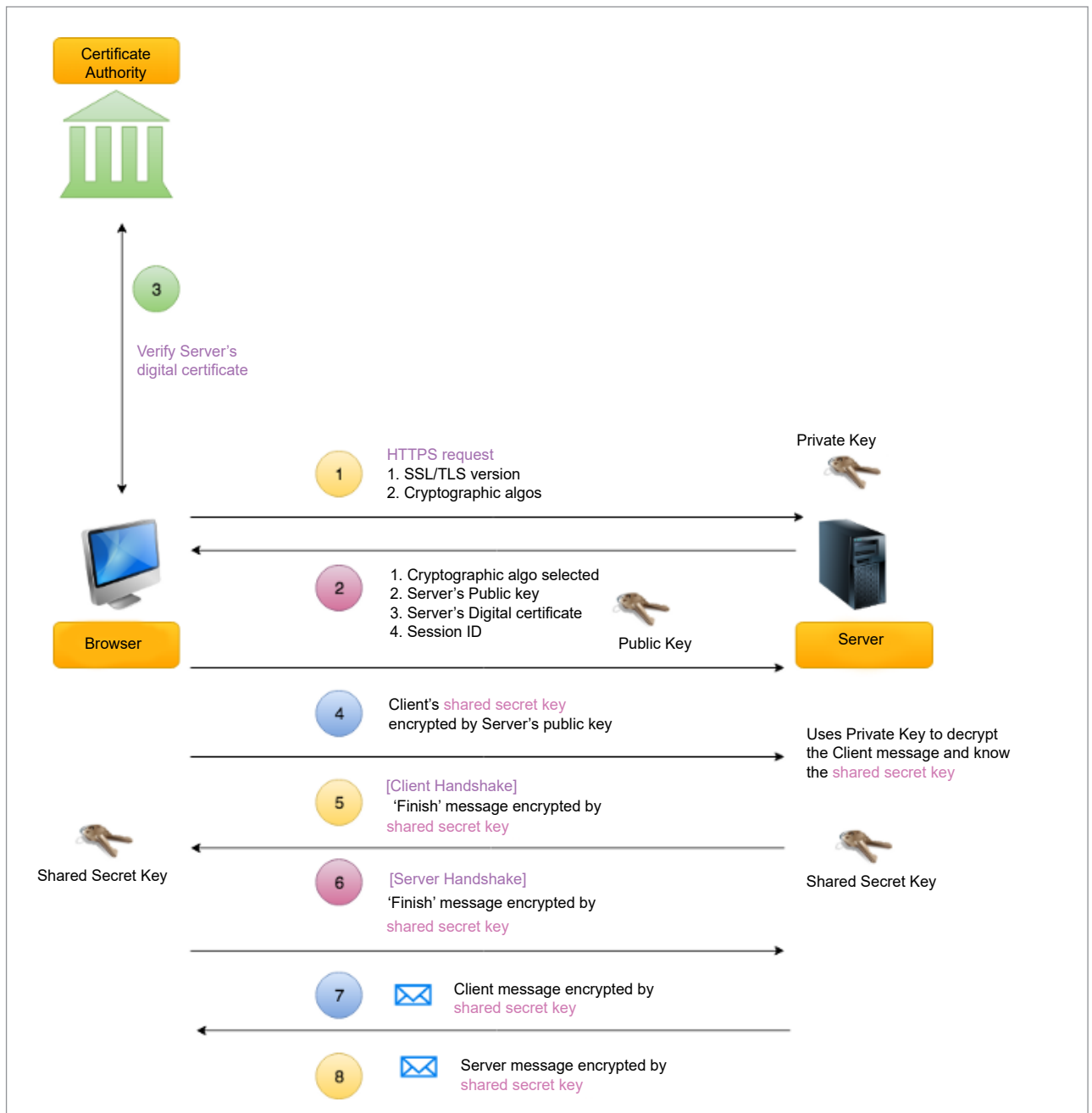


Figure 6: HTTP vs HTTPS/SHTTP (Credits: https://medium.com/@hiroki_ark/how-https-works-711e4828884)

key encryption.

HTTP vs HTTPS/SHTTP

HTTP is used for basic web connectivity; it occurs on port 80 and uses TCP in the transport layer. No security is provided by using HTTP, as the data is transmitted in plain text.

Security is enabled in the transport layer using SSL/TLS. The HTTPS protocol occurs on port 443. It is the same HTTP protocol but with security in the transport layer itself; a

different port number is used to differentiate it from HTTP.

The URL used for HTTP is `http://<servername>` and for HTTPS it is `https://<servername>`.

SHTTP or Secure HyperText Transfer Protocol is an alternative to HTTPS, but as it has not been adopted by many web browsers it's less common than HTTPS. Figure 5 explains the handshake between server and client.

MIME, POP, IMAP vs S/MIME, POP3S, IMAPS